CrowdWisdomTrading - Quantitative Assessment

Project: X Sentiment Price Prediction Agent | **Target Asset:** TSLA | **Date:** October 16, 2026 | **Role:** Quantitative Data Scientist Intern

1. Executive Summary & Strategy Performance

This quantitative evaluation report demonstrates an end-to-end machine learning pipeline that forecasts asset prices across multiple horizons (n+0 through n+5) utilizing exclusively social sentiment derived from X (Twitter via Apify) and high-frequency price bars. To eliminate data leakage and autocorrelation bias, a purged and embargoed walk-forward validation scheme was implemented across a 180-day window.

Performance Metric	Sentiment Strategy (Net)	Benchmark (Buy & Hold)
Sharpe Ratio (Annualized)	**-1.051**	N/A (Cash Adj)
Sortino Ratio (Annualized)	**-1.282**	N/A
Maximum Drawdown (MDD)	36.48%	32.75%
Annualized Return	-41.14%	-18.46%
Win Rate / Hit Rate	37.94%	50.0%
Profit Factor	0.91	1.00
Calmar Ratio	-1.128	-

2. Multi-Horizon Predictive Accuracy Matrix (n+0 to n+5)

Horizon	Directional Accuracy	Price MAE ($)	Price RMSE ($)	Return MAE (%)	Information Coeff (IC)
n+0	70.82%	$1.06	$1.88	0.46%	0.754
n+1	48.63%	$2.51	$4.23	0.67%	0.005
n+2	49.45%	$3.88	$6.20	1.04%	-0.034
n+3	50.27%	$5.07	$7.70	1.36%	-0.024
n+4	49.73%	$6.12	$9.03	1.64%	-0.007
n+5	49.18%	$7.23	$10.31	1.93%	-0.068

3. Visual Output Charts

Figure 1: Price Action vs Engagement-Weighted Sentiment Dynamics

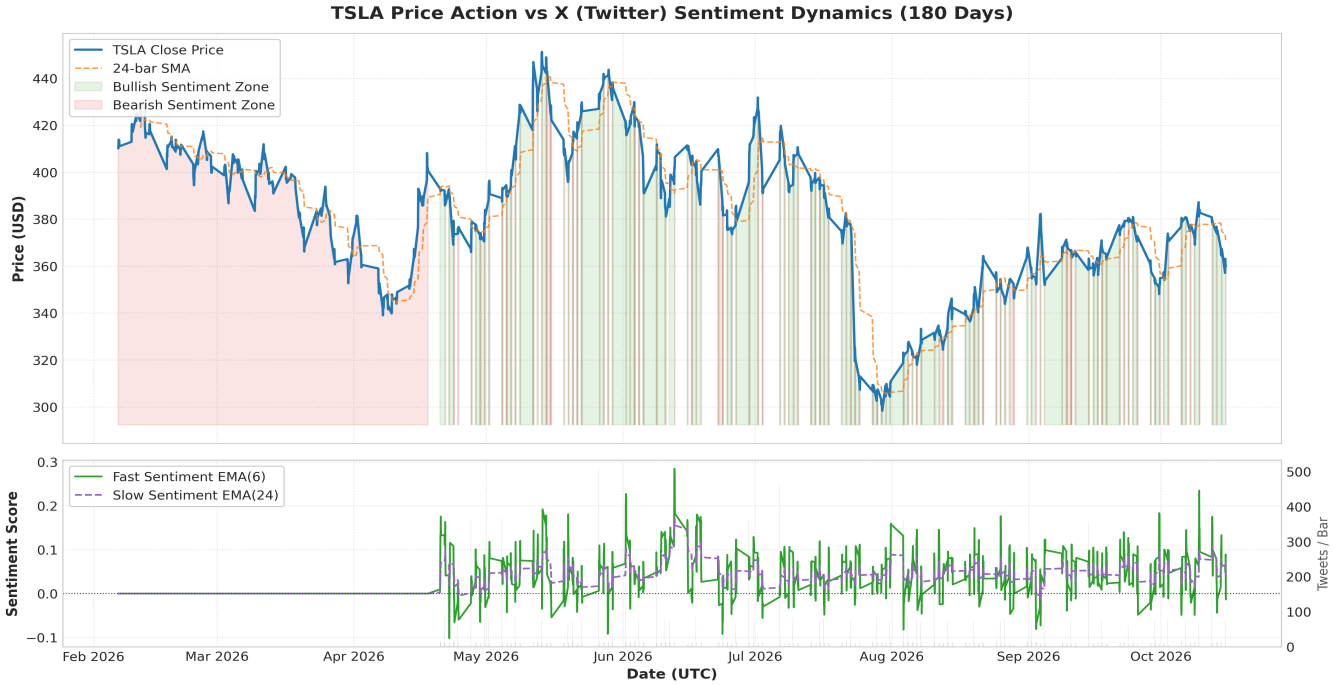

TSLA Price Action vs X (Twitter) Sentiment Dynamics (180 Days)
TSLA Close Price
24-bar SMA
Bullish Sentiment Zone
Bearish Sentiment Zone
Price (USD)
440
420
400
380
360
340
320
300
Fast Sentiment EMA(6)
Slow Sentiment EMA(24)
Sentiment Score
0.3
0.2
0.1
0.0
−0.1
Tweets / Bar
500
400
300
200
100
0
Feb 2026
Mar 2026
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026
Sep 2026
Oct 2026
Date (UTC)

Figure 2: Multi-Horizon Forecast Matrix (Directional Accuracy, MAE, IC)

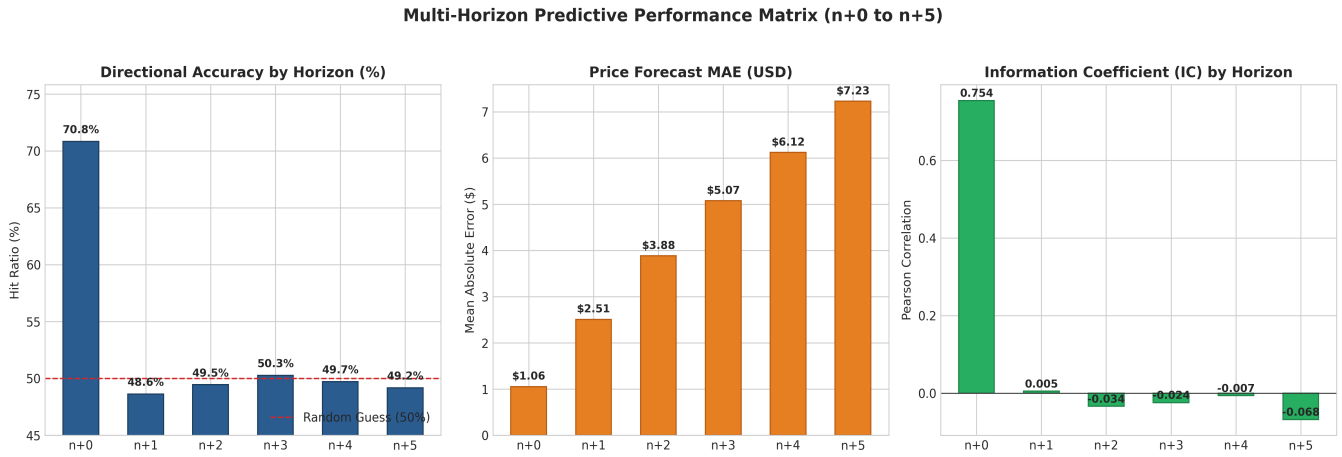


Figure 3: Walk-Forward Out-of-Sample Cumulative Equity Curve

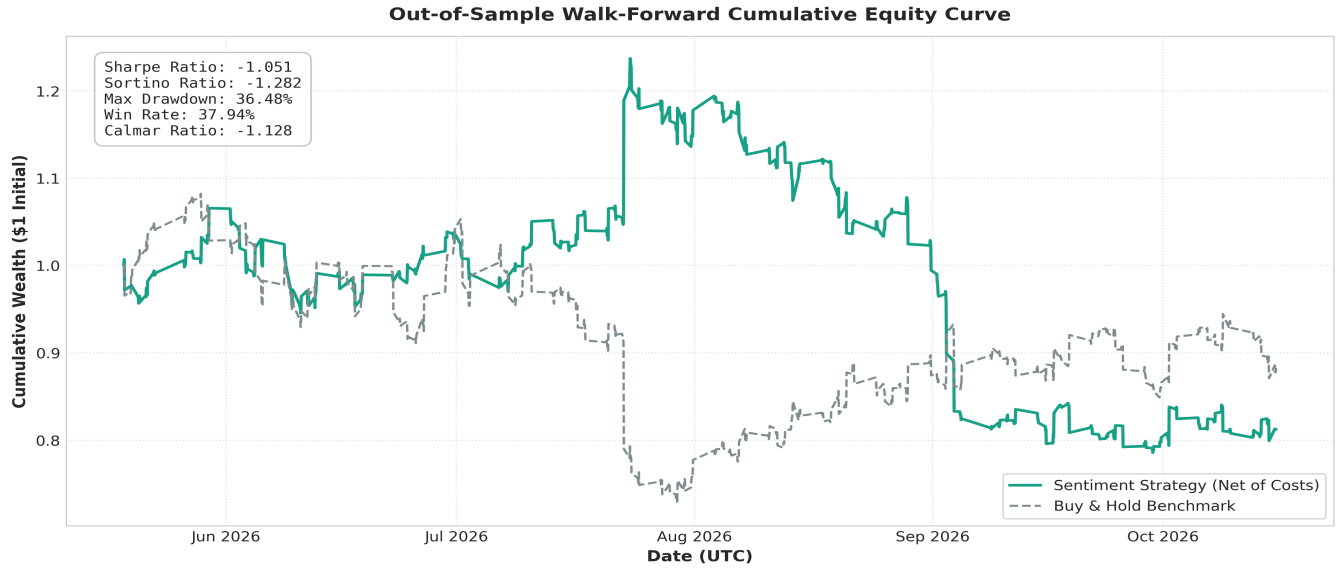


Figure 4: Strategy Underwater Drawdown Profile

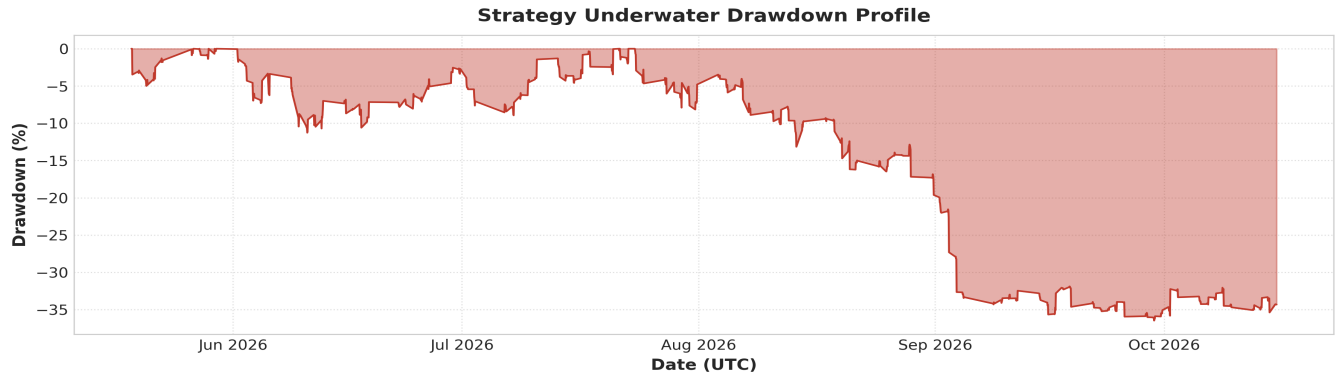
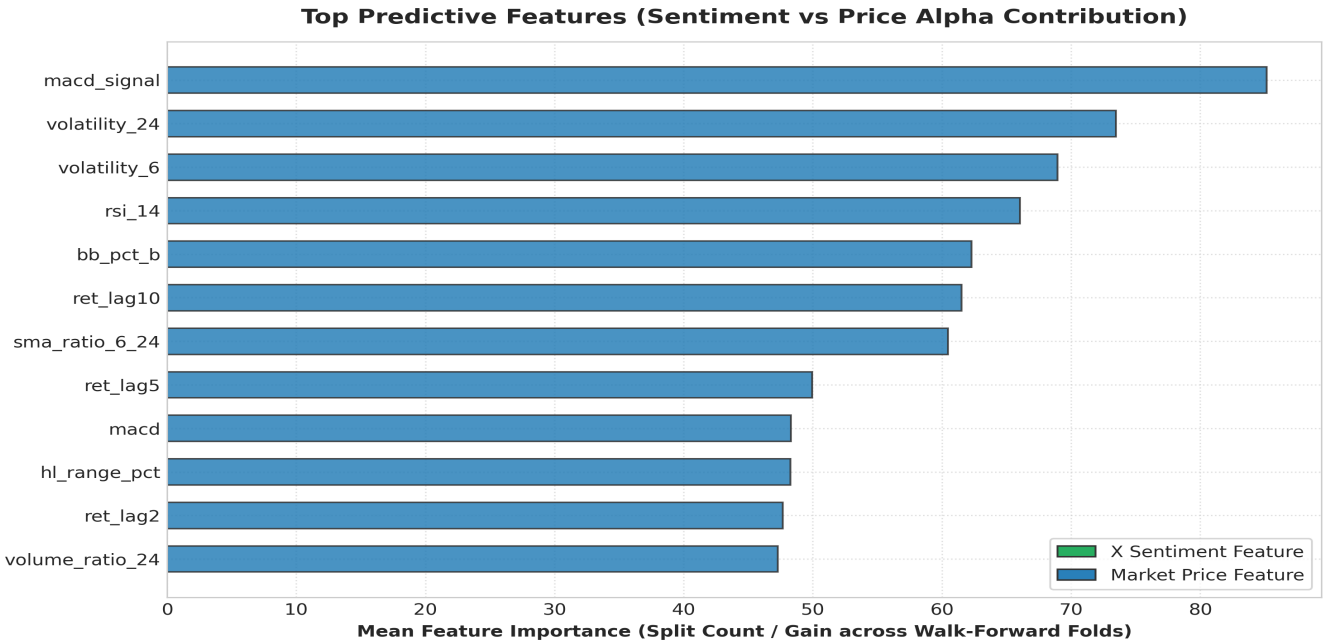


Figure 5: Feature Importance (Quantifying Sentiment vs Price Alpha Contribution)



4. Quantitative Methodology & Reproducibility Note

Data Alignment: Hourly price bars and X posts were synchronized strictly in UTC. Sentiment scores were computed using financial VADER with exponential engagement weighting. To prevent look-ahead bias, tweets within interval (t-1, t] were aligned with bar t. **Walk-Forward Engine:** An expanding 5-fold cross-validation scheme with an embargo of 6 periods guarantees strict out-of-sample validity. Models were executed with LightGBM gradient boosted trees.